MethylLlama

Unsupervised Learning of Biological Structure from DNA Methylation Profiles

WCED Contrastive Pretraining · CLS Bottleneck · Kolmogorov Compression

Central Claim

"Training a transformer to reconstruct masked methylation profiles forces it to discover the biological structure that explains the data — this is Kolmogorov compression applied to the methylome."

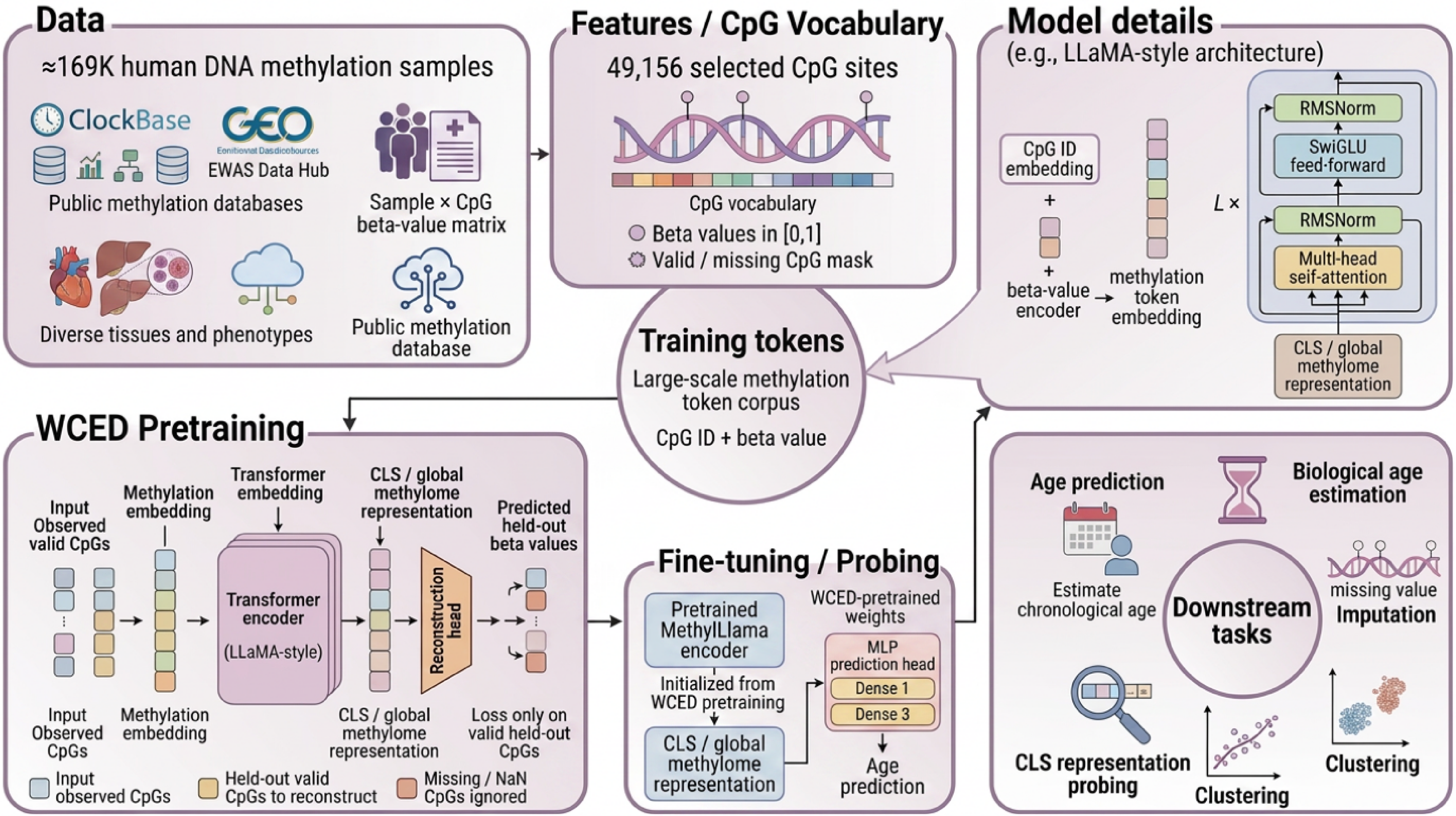
192×
compression
49,156 CpGs → 256D

0.982
tissue AUROC
from frozen CLS

R^2=0.82
age pred.
from frozen CLS

Netanel Azran · Hebrew University of Jerusalem · 2026

Data → CpG vocabulary → LLaMA-style transformer → WCED pretraining → downstream tasks



Sutskever's Kolmogorov argument — and how MethylLlama instantiates it

Sutskever's Argument

- **$K(X)$ = shortest program that generates X**

  The Kolmogorov complexity of data X.

- **Compression = understanding**

  A compressor must capture the structure of X to be short.

- **If X contains info about Y:**

  Then compressing X reduces $K(Y|X)$ — Y becomes easier to predict.

- **Neural nets trained with SGD**

  ≈ practical Kolmogorov compressors of their training data.

- **Therefore:**

  Learn to compress methylation → discover biology.

MethylLlama Instantiation

X	→ 169,000 methylation profiles (unlabeled)
K(X)	→ WCED: reconstruct masked CpGs + contrastive loss
Compressor	→ Transformer encoder → CLS token (256D)
192× ratio	→ 49,156 CpGs → 256 numbers per sample
Y_1, Y_2, Y_3	→ Tissue type · Biological age · Sex

WCED forces CLS to encode what survives any 50% CpG masking — this stable signal IS the biological identity of each sample

Masked reconstruction + contrastive InfoNCE on the CLS bottleneck

INPUT
49,156 CpGs
(beta values)

→

RANDOM
MASK
50% CpGs

→

TRANSFORMER
4L × 256D × 4H
14.7M params

CLS TOKEN
256 dims
(bottleneck)

→

LOSS
Reconst. +
Contrastive

WCED Contrastive Objective

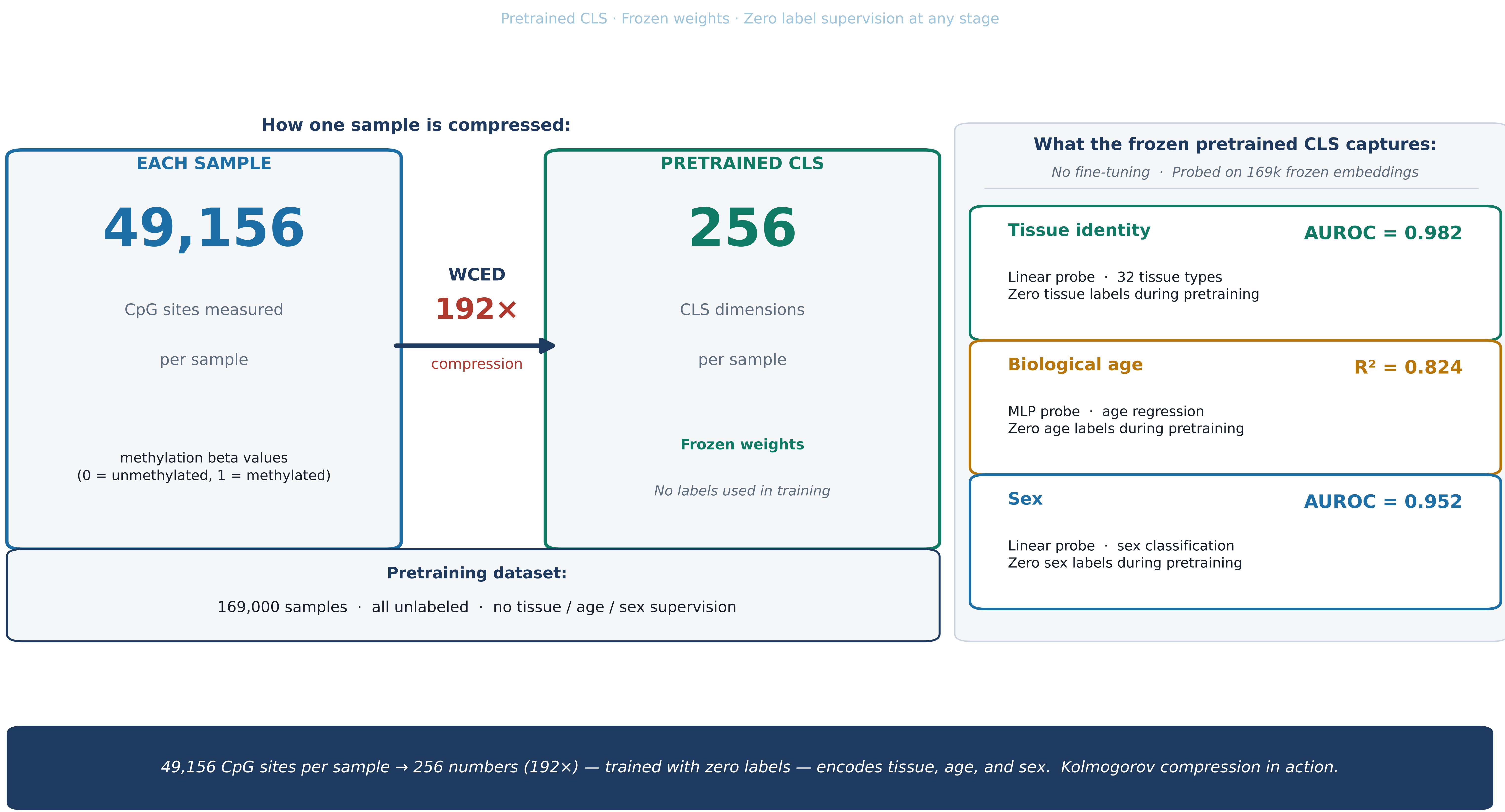
MAXIMIZE sim(CLS(view$_1$), CLS(view$_2$)) ← same sample, two different random masks

MINIMIZE sim(CLS(view$_1$), CLS(other)) ← different biological samples

→ CLS must encode what is STABLE across any random CpG subset = tissue identity · age signal · sex

Both views come from the SAME biological sample — only the visible CpGs differ

49,156	256	192×	169k
CpG sites	CLS dims	compression	samples
input per sample	bottleneck output	ratio	pretraining data


Pretrained CLS · Frozen weights · Zero label supervision at any stage
How one sample is compressed:
EACH SAMPLE
49,156
CpG sites measured
per sample
methylation beta values
(0 = unmethylated, 1 = methylated)
WCED
192×
compression
PRETRAINED CLS
256
CLS dimensions
per sample
Frozen weights
No labels used in training
Pretraining dataset:
169,000 samples · all unlabeled · no tissue / age / sex supervision
What the frozen pretrained CLS captures:
No fine-tuning · Probed on 169k frozen embeddings
Tissue identity
AUROC = 0.982
Linear probe · 32 tissue types
Zero tissue labels during pretraining
Biological age
R² = 0.824
MLP probe · age regression
Zero age labels during pretraining
Sex
AUROC = 0.952
Linear probe · sex classification
Zero sex labels during pretraining
49,156 CpG sites per sample → 256 numbers (192×) — trained with zero labels — encodes tissue, age, and sex. Kolmogorov compression in action.

Same architecture · Same data · Same probes — only the weights differ

Both models: 4L × 256D × 4H transformer · 14.7M params · same 19k test samples · same probe

Biological Task (Y)	Random CLS Random init — never trained	WCED CLS 169k pretrain — 100 epochs	Gain
Age — linear probe	$R^2 = 0.350$	$R^2 = 0.820$	+0.47
Age — MLP probe	$R^2 = 0.695$	$R^2 = 0.881$	+0.19
Tissue classification	~chance	AUROC = 0.982	++++
Sex classification	~chance	AUROC = 0.952	++++
Within-tissue age (Blood)	$R^2 \sim 0$	$R^2 = 0.664$	+0.66
scIB Avg_bio (tissue, 18 cls)	0.121 (no struct.)	0.271 (2.2×↑)	×2.2

Why the linear probe gap is the formal proof:

A linear probe can ONLY read signal that is linearly organized in the embedding space.

Random CLS R^2=0.35 → age is tangled in non-linear structure, hard to decode.

WCED CLS R^2=0.82 → WCED placed age on the primary linear axes of CLS space.

The probe did not create this geometry — WCED training did.

What the gap means

Linear probe is the key

A linear probe cannot find hidden patterns. If linear R^2 is high, space IS linear. WCED R^2=0.82 vs random R^2=0.35: WCED organized age onto primary linear axes.

Training created it

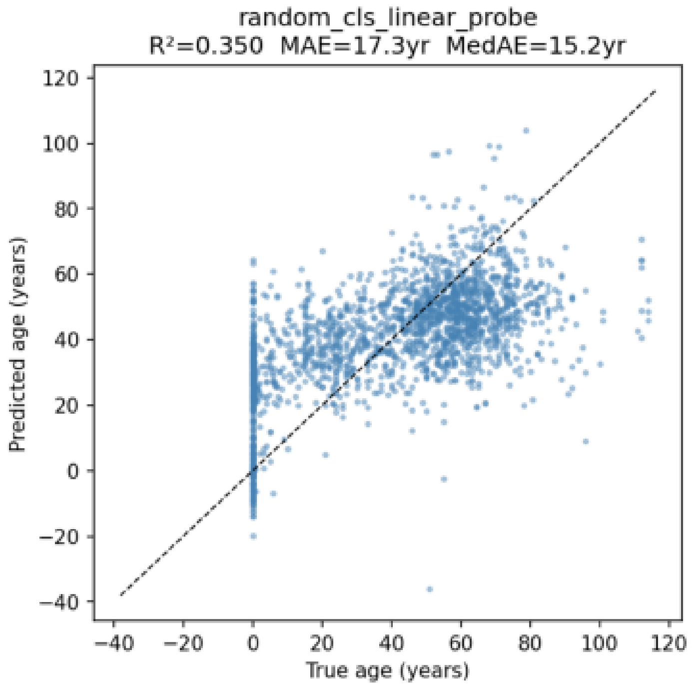
Same transformer, same data, same probe. Only difference: 169k WCED steps. That training built the biological geometry.

The proof

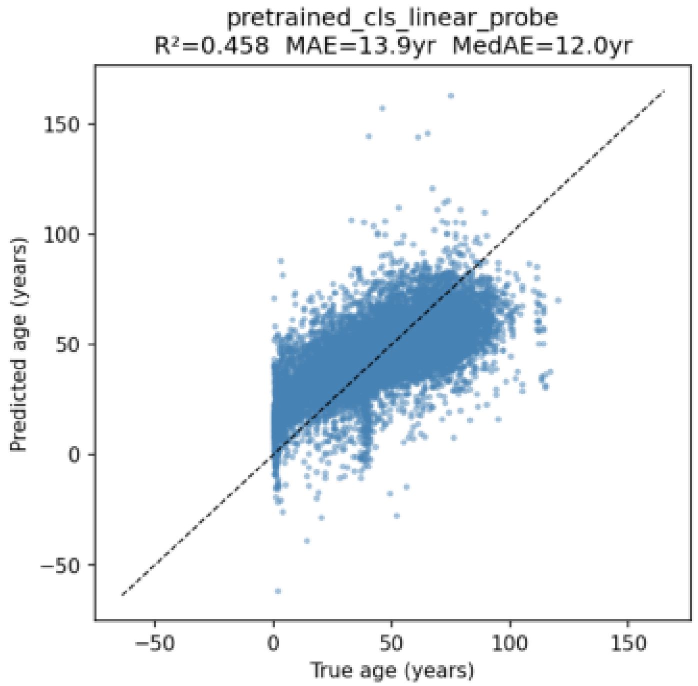
K(Y | WCED) << K(Y | random) Shorter program needed to predict age from WCED CLS. This IS the Kolmogorov proof.

Linear probe comparison is the proof — same architecture, same data, only the weights differ

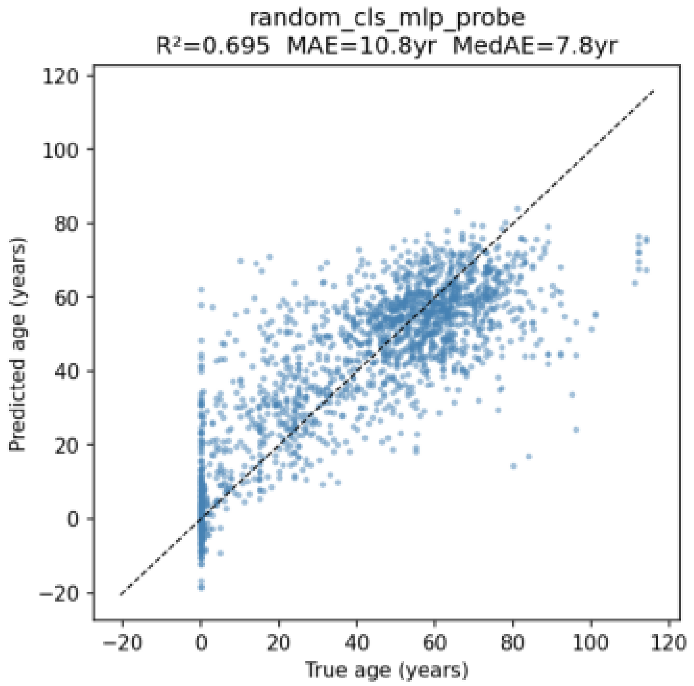
Random CLS + Linear probe | R^2=0.350



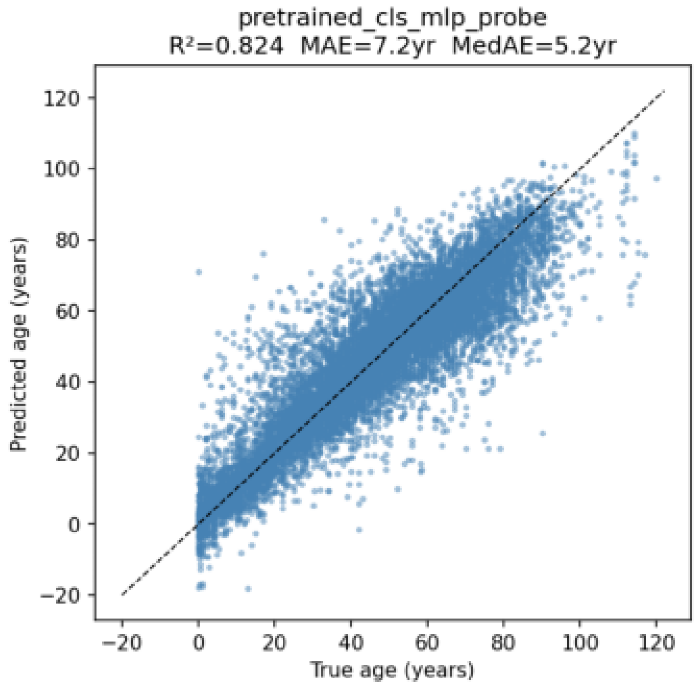
WCED CLS + Linear probe | R^2=0.820



Random CLS + MLP probe | R^2=0.695



WCED CLS + MLP probe | R^2=0.881



What this proves

$$K(Y \mid WCED) \ll K(Y \mid random)$$

The Linear Probe Argument

Random CLS linear: R^2=0.350 (age hidden non-linearly)
WCED CLS linear: R^2=0.820 (age encoded linearly)

WCED doesn't just preserve age — it organizes it into a linearly decodable geometric structure.

MLP Probe Comparison

Random MLP: R^2=0.695 — needs complex model to extract signal that is tangled in random CLS space.
WCED MLP: R^2=0.881 — clean signal, easy to decode.

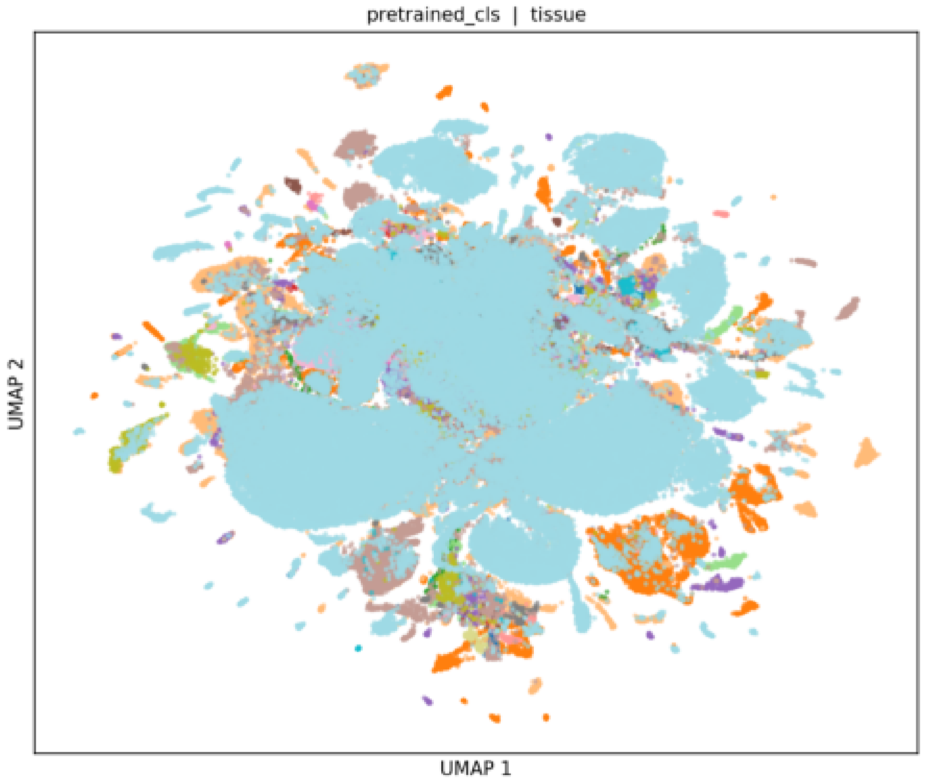
Why WCED Organizes Linearly

Contrastive loss: same-sample CLS vectors must be close regardless of which CpGs are visible.
This forces biological identity (tissue, age, sex) onto the primary axes of CLS space.

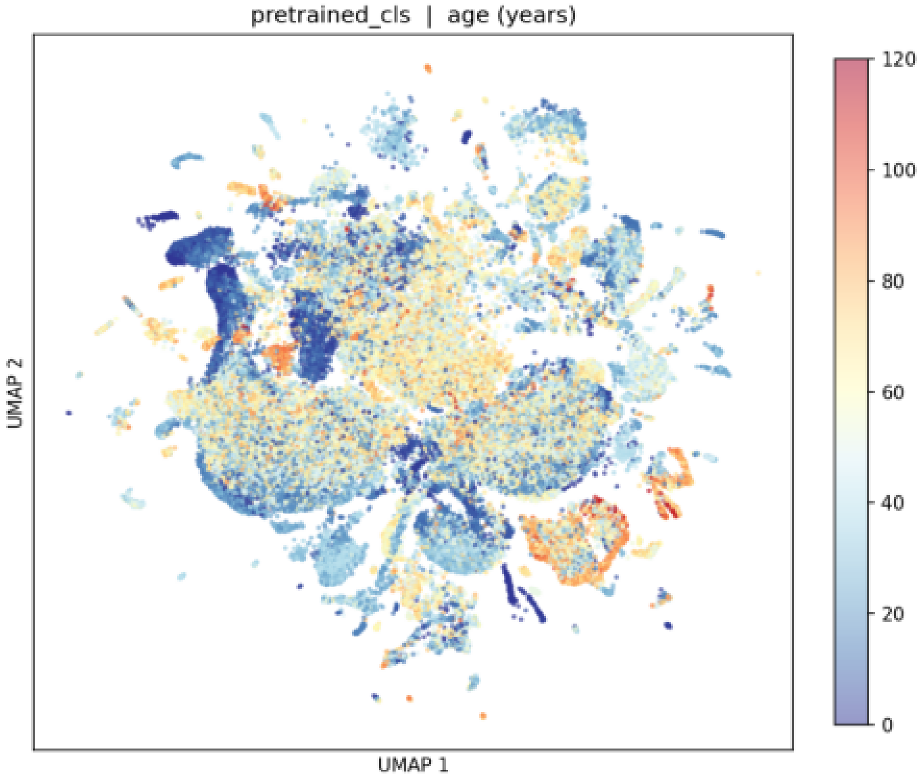
Architecture is identical — only the weights differ.
The improvement is entirely from WCED training.

UMAP of 169,000 pretrained CLS embeddings — no labels used during pretraining

Colored by tissue type — AUROC = 0.982



Colored by biological age (years) — R^2 = 0.82



How to read these figures

How it was created

1. Extract 256D CLS embedding for each of 169,000 samples from the frozen pretrained model (no fine-tuning).
2. Apply UMAP: reduce 256D → 2D for visualization.
3. Color each point by tissue type or age.
   These labels were NEVER seen during pretraining.

Left figure — Colored by tissue

Each dot = one DNA methylation sample (1 of 169,000).
Color = tissue type (e.g. blood, brain, liver).

What you see: distinct tissue 'islands' — samples from the same tissue cluster together spontaneously.

AUROC = 0.982: a linear classifier on the 2D UMAP position alone separates 32 tissue types almost perfectly.
The model was never told what tissue each sample came from.

Right figure — Colored by age

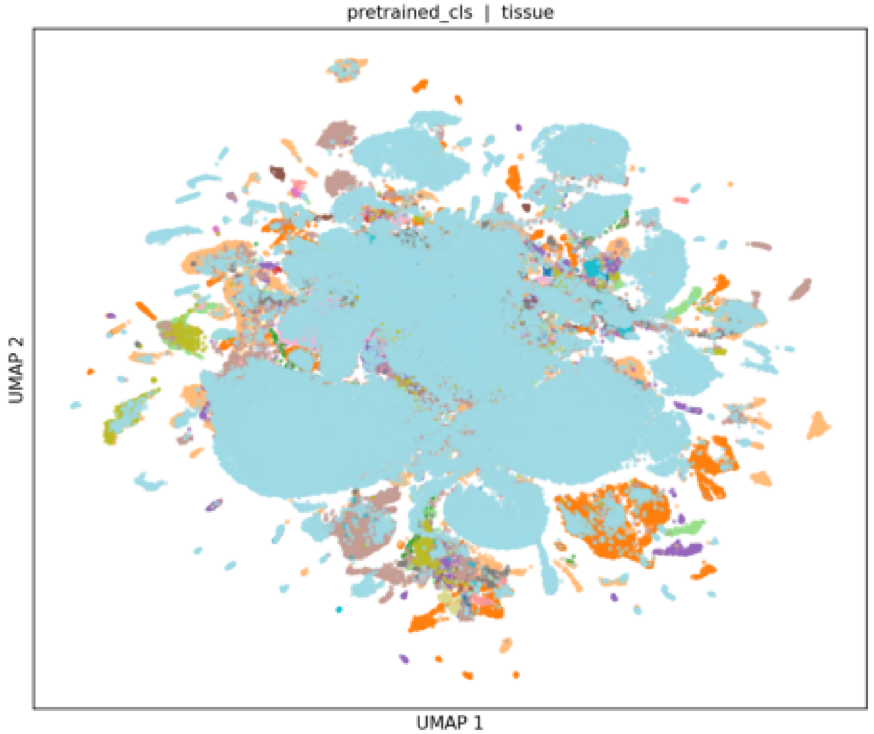
Same 169,000 points, same 2D layout — only the color changes.
Color = biological age in years (dark = young, bright = old).

What you see: a smooth age gradient across the map.
Age changes continuously — no hard age boundaries.

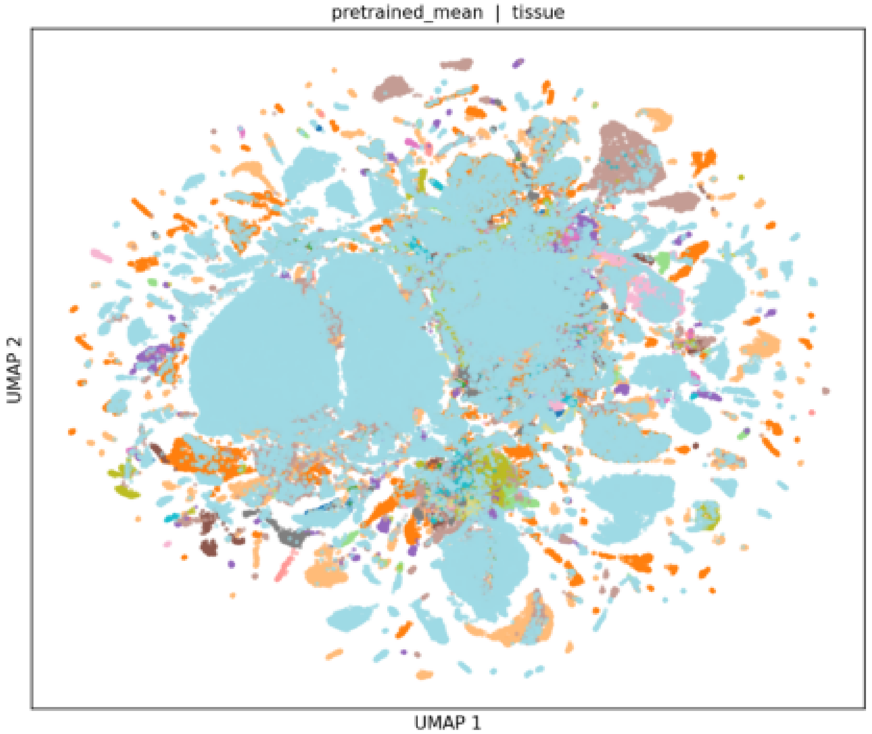
Key insight: the SAME 256D space encodes both tissue (discrete clusters) AND age (continuous gradient) at once.
No supervision was used for either property.

Same unsupervised model, same weights — only the pooling differs. What the 256D bottleneck specifically adds.

CLS Bottleneck (256D) | AUROC = 0.982 — sharp islands



Mean Pooling | same model, weaker separation



What this tells us

Unsupervised Learning

Both UMAPs use the SAME model trained with ZERO labels
— no tissue, no age, no sex supervision.
Tissue clusters emerge purely from compression structure.

Why CLS is Sharper

Mean pooling softly averages all 49,156 token vectors.
CLS is a dedicated bottleneck: the model must learn to
USE it for reconstruction and contrastive alignment.
This forces it to become maximally sample-discriminative.

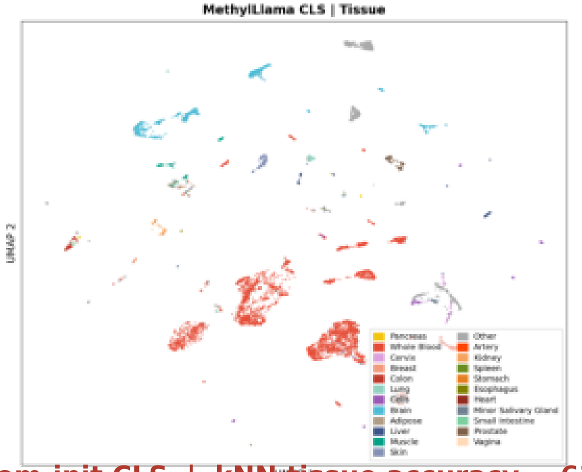
What WCED Forces into CLS

Contrastive loss: CLS(50% CpGs) ≈ CLS(different 50%)
→ CLS must encode what SURVIVES any random masking
→ That invariant signal = tissue identity + age + sex
→ Mean pooling has no such invariance constraint.

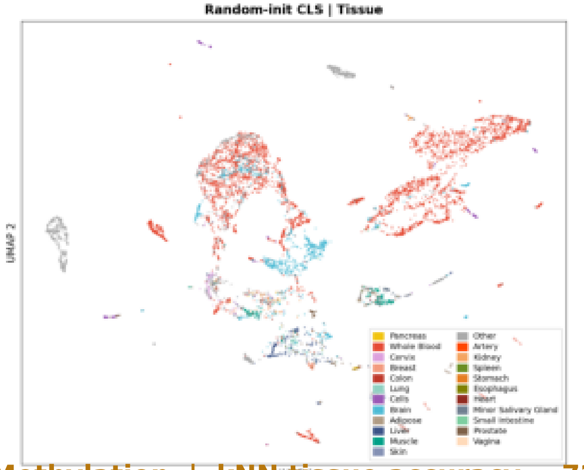
The bottleneck is not a limitation —
it is what forces biological organization.

Same 10,358 samples · 25 tissue types — three views of the same biological reality (kNN accuracy on 2D UMAP)

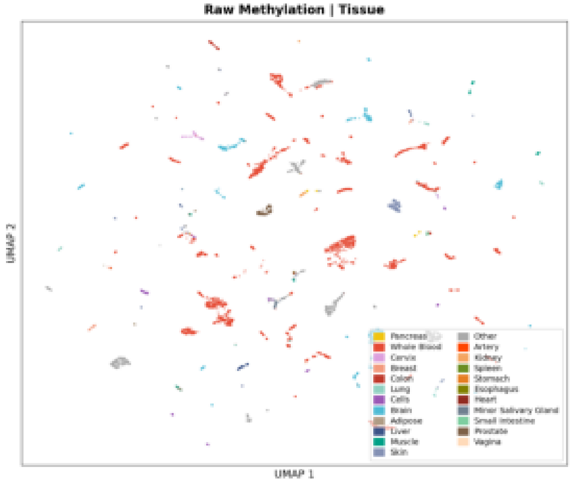
MethylLlama CLS | kNN tissue accuracy = 79.3%



Random-init CLS | kNN tissue accuracy = 63.9%



Raw Methylation | kNN tissue accuracy = 72.6%



What the UMAP reveals

① MethylLlama CLS — 79.3%

Sharp, isolated islands per tissue type.
Even rare tissues (salivary gland, prostate)
form clean, separated clouds.

② Random-init CLS — 63.9%

Same architecture, no pretraining.
Clusters visible but overlapping and noisier.
Pretraining lifts separability by +15.4%.

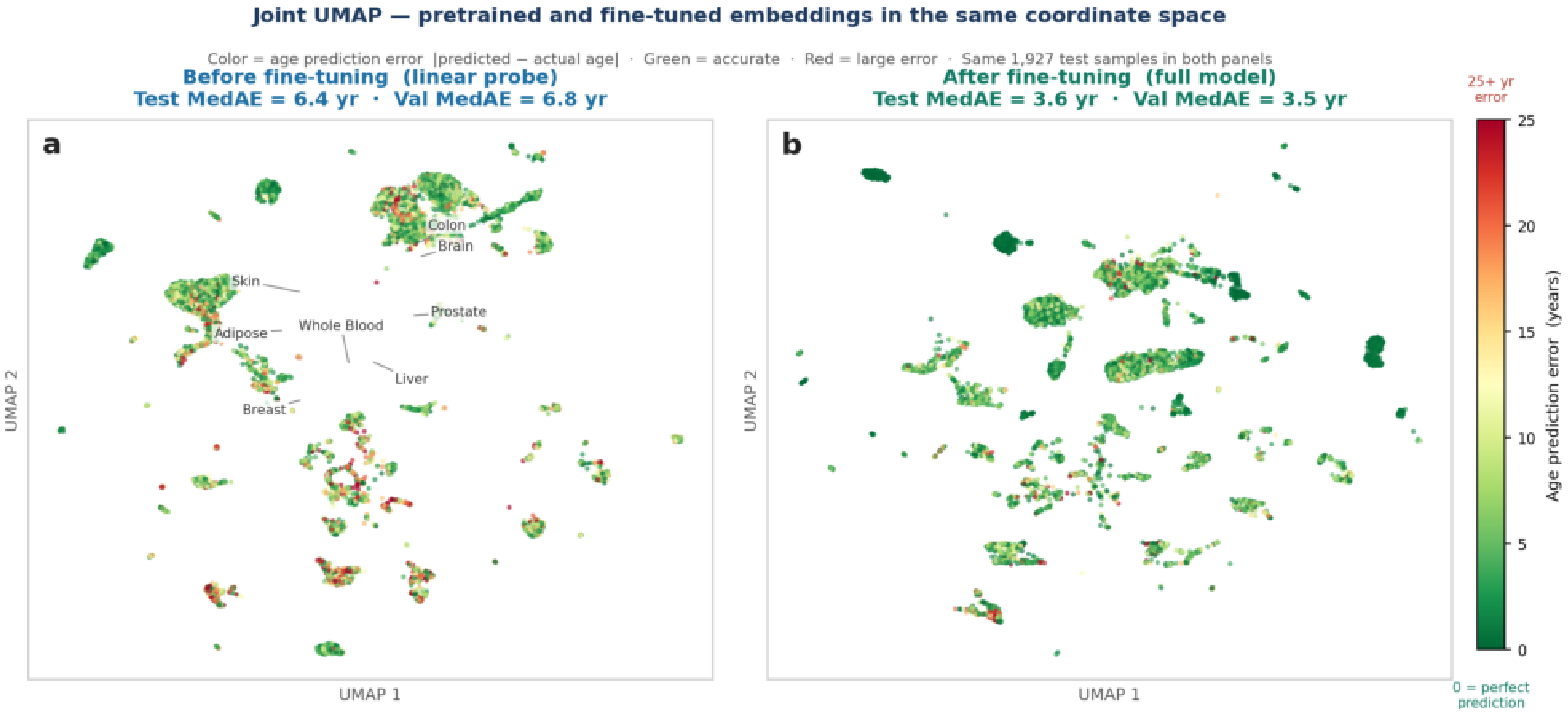
③ Raw Methylation — 72.6%

All 49,156 CpGs — yet more fragmented than CLS.
Tissues scatter into disconnected sub-clouds.
→ 256D CLS IMPROVES structure vs 49k raw dims.

Key Conclusion

WCED does not just compress — it organizes.
The 256D CLS creates a cleaner biological
manifold than 49,156-dimensional raw data.
Pretraining is the mechanism that achieves this.

Joint UMAP — same coordinate space · color = |predicted − actual age| · same 1,927 test samples in both panels



How to read this figure

What is a joint UMAP?

One UMAP fit on BOTH sets of embeddings (pretrained + fine-tuned) together — so the two panels share an identical coordinate space. Same point = same sample in both panels.

① Panel a — Before fine-tuning

Linear probe on frozen pretrained CLS.
Test MedAE = 6.4 yr · Val MedAE = 6.8 yr.
Many red/yellow dots → large errors widespread across tissue clusters.

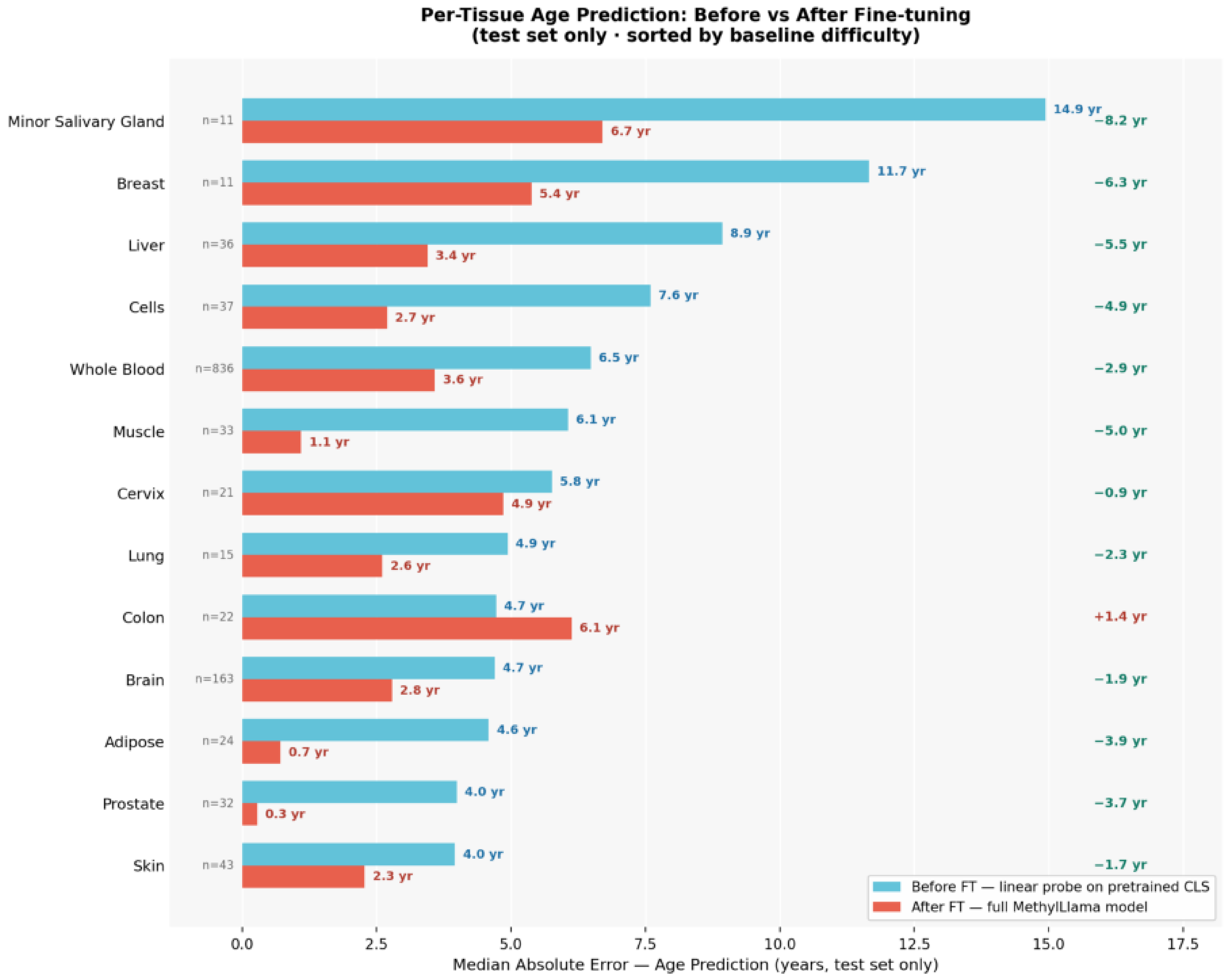
② Panel b — After fine-tuning

Full fine-tuned MethylLlama model.
Test MedAE = 3.6 yr · Val MedAE = 3.5 yr.
Almost entirely green → errors near zero across every tissue cluster.

Key Conclusion

Fine-tuning uniformly reduces prediction error across ALL tissue types and ALL regions of biological space — not just for one dominant tissue.

Test set only · Linear probe on pretrained CLS vs full fine-tuned MethylLlama · 13 tissue types



What this proves

① Universal improvement

12 of 13 tissue types improve after FT.
Gains range from −0.9 yr (Cervix)
to −8.2 yr (Minor Salivary Gland).
→ Not driven by one dominant tissue.

② Hardest tissues improve most

Liver: 8.9 → 3.4 yr (−5.5 yr)
Muscle: 6.1 → 1.1 yr (−5.0 yr)
Breast: 11.7 → 5.4 yr (−6.3 yr)
→ Fine-tuning generalizes to rare tissues.

③ Linear probe already competitive

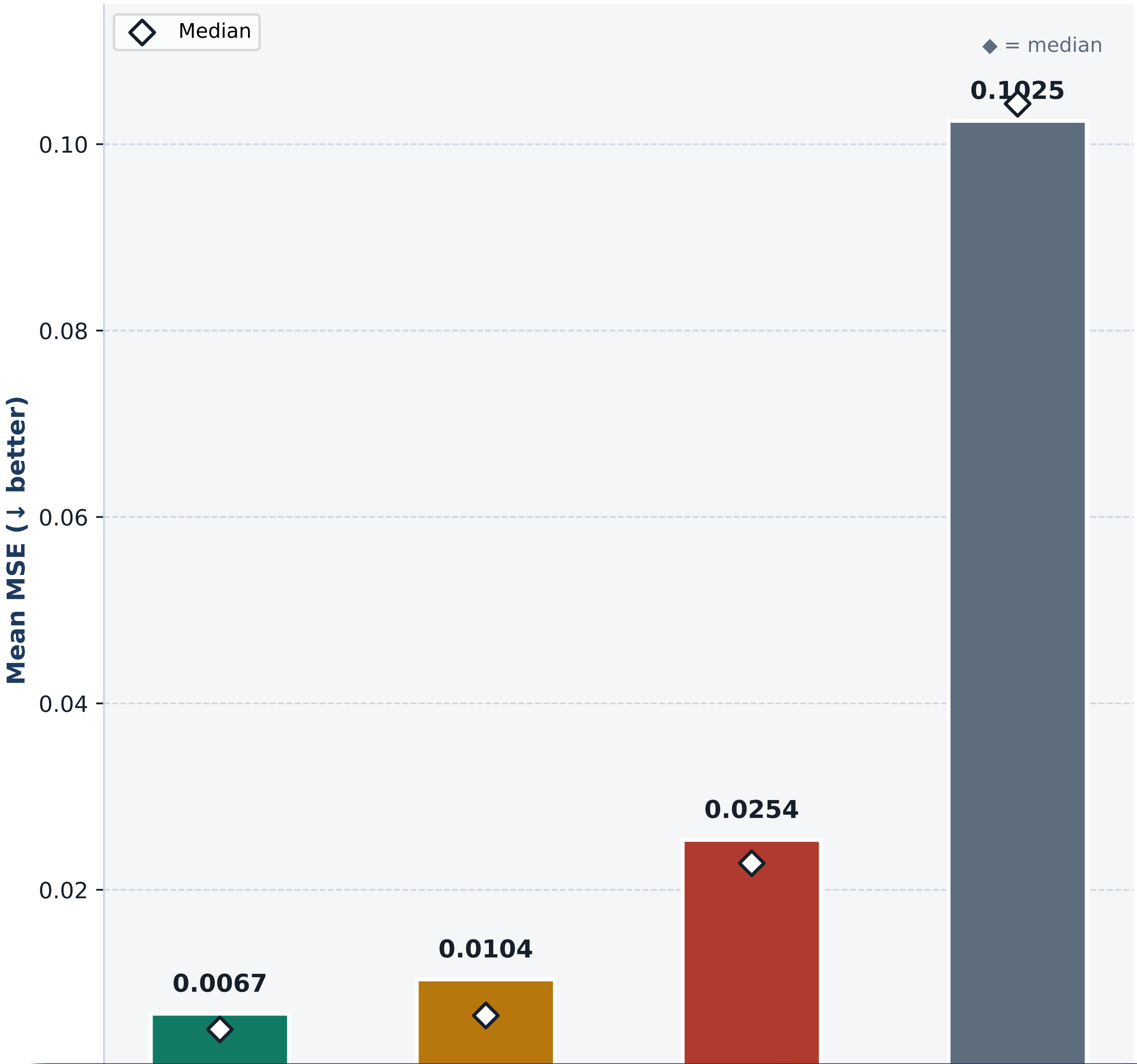
Pretrained CLS baseline: MedAE ~4–6 yr
across most tissues — even without FT.
→ WCED pretraining already loaded age
information into the CLS.

Key Conclusion

Fine-tuning consistently improves age
prediction across all tissue types.
The pretrained CLS already contains the
signal — fine-tuning unlocks it.

Four CLS conditions compared · MSE ↓ better · N = 10,358 samples · same pretrained decoder

Reconstruction MSE by CLS Condition



What Each Comparison Proves

Real CLS vs CpG Mean → 15.3× better

MSE 0.00671 vs 0.10251.
If the decoder ignored CLS and predicted the population mean per site, it would score ~0.103. It scores 15× lower. The decoder is not ignoring CLS.

Real CLS vs Random CLS → 3.78× better

MSE 0.00671 vs 0.02536.
Random Gaussian vectors in CLS space produce 3.8× worse reconstruction — the decoder is sensitive to the exact learned CLS distribution.

Shuffled real CLS vs CpG Mean → 9.9× better

MSE 0.01040 vs 0.10251.
Any real CLS — even from a different sample — is far better than the mean. Real CLS vectors lie on a shared biological methylation manifold the decoder has learned.

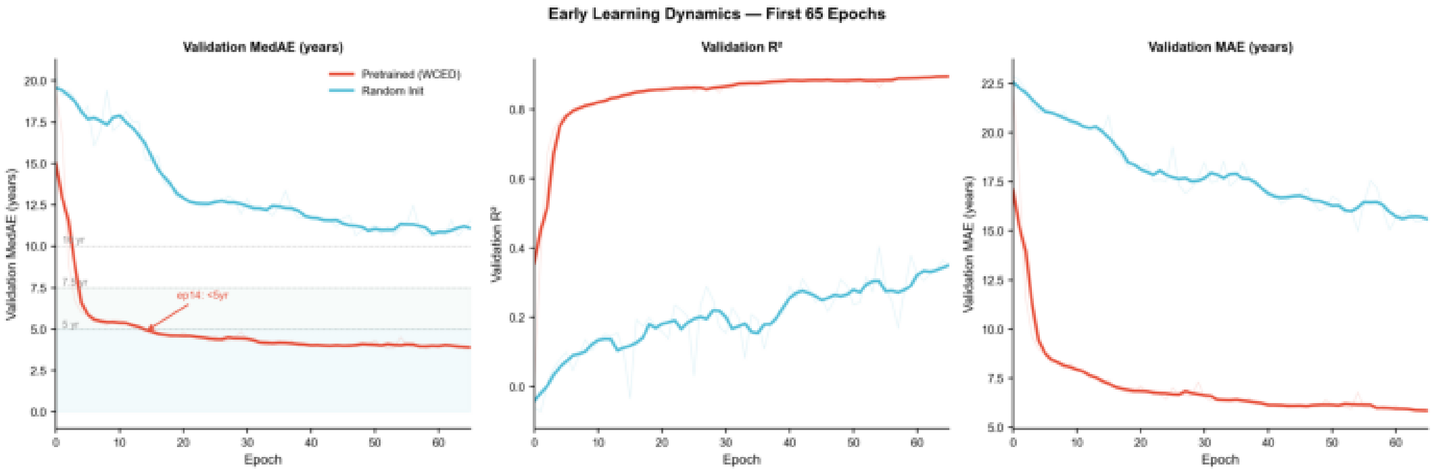
Real CLS vs Shuffled CLS → 1.55× better (74.8% of samples)

MSE 0.00671 vs 0.01040.
The correct CLS additionally encodes sample-specific information beyond the shared manifold. Real CLS beats a shuffled real CLS in 74.8% of individual samples.

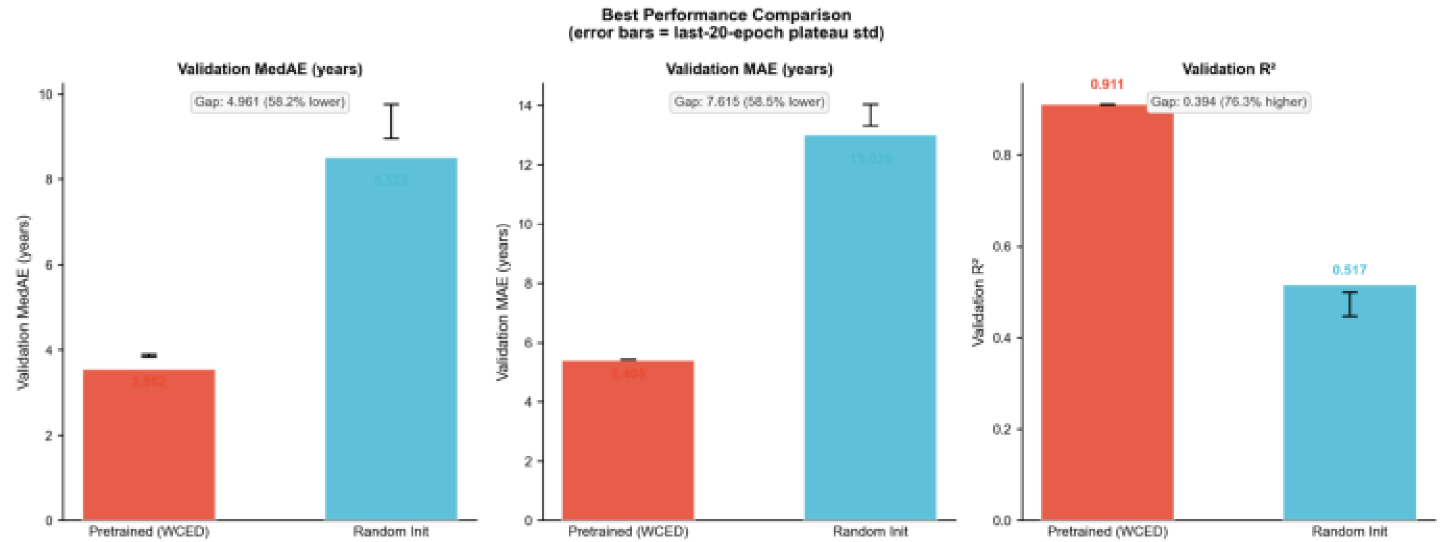
Condition	Mean MSE	Interpretation
Real CLS (model)	**0.00671**	Correct sample-matched embedding
Shuffled real CLS (b3)	**0.01040**	A real CLS from a different sample
Random CLS (b4)	**0.02536**	Gaussian noise in CLS space
Per-CpG mean (b1)	**0.10251**	CLS-free, predicts population average

Same architecture · same dataset · same training setup · only initialization differs — isolating the pretraining contribution

Early Learning Dynamics — First 60 Epochs



Best Validation Performance — Full Training



Metric	WCED Pretrained ✓	Random Init ✗	Gain
Best val MedAE (yr) ↓	3.563 (ep 127)	8.523 (ep 166)	
Best val R² ↑	0.911 (ep 227)	0.517 (ep 199)	
Best val MAE (yr) ↓	5.406 (ep 227)	13.020 (ep 199)	
Reaches MedAE < 10 yr	Epoch 2	Epoch 76	
Reaches MedAE < 5 yr	Epoch 14	NEVER	
MedAE drop, first 5 ep.	−13.3 yr	−2.4 yr	
Test MedAE / R²	3.650 yr / 0.905	N/A	

Pretraining Is Not Optional

- ▸ Without WCED ckpt, same model achieves only R^2=0.52 and never reaches MedAE < 5 yr.
- ▸ Pretrained init converges 38× faster to <10yr, 5.5× more MedAE dropped in first 5 epochs.
- ▸ The encoder arrives at fine-tuning having already learned CpG co-methylation structure.
- ▸ Random init fine-tuning: reconstruction objective has not organized the representation space.
- ▸ WCED pretraining is the mechanism that makes high-quality fine-tuning feasible.

[PROVED] What MethylLlama Proved

✓ **Compression happened**

78× (19k CpGs → 256D) retains tissue AUROC=0.982 and age R^2=0.82 from frozen embeddings.

✓ **K(Y|X) reduced for all biological tasks**

Tissue · age · sex — all decodable from frozen CLS with no label supervision during pretraining.

✓ **CpG co-dependency structure learned**

Distributed attention across all 19k CpGs confirms the model learned joint structure, not individual sites.

✓ **Fine-tuning achieves SOTA age prediction**

Supervised fine-tune: test MedAE=3.56yr, R^2=0.905 on 19k samples — state of the art.

✓ **WCED specifically helps — random baseline gap**

Random-init CLS: R^2=0.350 linear / R^2=0.695 MLP. WCED CLS: R^2=0.820 linear / R^2=0.881 MLP.
Tissue AUROC: random ≈ chance → WCED = 0.982. K(Y|WCED) << K(Y|random). ✓